
LLM-Powered Automation of a Dark Matter Constraint Repository

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Abstract

Dark matter constraint repositories are critical community infrastructure, giving experimentalists and theorists a shared landscape of existing bounds. Yet the most widely-used repositories are maintained by individual volunteers, creating a sustainability risk as the pace of new results accelerates. We present a large language model (LLM) pipeline that monitors arXiv, extracts limit curves from papers, integrates them as code, and opens pull requests (PRs) for human review. On a 346-paper benchmark whose ground truth is the upstream-curated repository itself, the pipeline classifies the coupling type correctly for 90.5% of papers and reaches a median coupling residual of 0.33 dex (a factor of two for 48% of curves), with 76% mean mass-range coverage. Three design choices drive this: (i) treating each extraction as a noisy sample and reconciling several reads by a consensus vote with source-quality tiering; (ii) a convention-canonicalization layer whose per-file conversion factors were derived with a guided physics-derivation assistant (GPD), code-verified and citation-audited; and (iii) a scoring methodology that separates genuine extraction error from non-comparability. The remaining difficulty is concentrated in rare coupling types with idiosyncratic conventions (macro-averaged residual 1.1 dex). The pipeline is deployed and has generated limit proposals; none have merged — governance of AI-generated scientific data is itself an unsolved problem.

1 Introduction

The experimental search for ultralight dark matter is undergoing a “renaissance” [1], with new proposals targeting masses from 10^{-22} eV to several eV using techniques from quantum-limited cavities to dish antennas and nuclear magnetic resonance [2, 3]. The result is an ever-denser landscape of constraints across at least 15 distinct *coupling types* (ways a dark matter candidate can interact with Standard Model particles, e.g. axion–photon or dark photon kinetic mixing) [4], each with its own conventions for units, parameterisations, and confidence levels.

Dark matter constraint repositories compile these bounds into publication-quality *exclusion plots* — diagrams showing which regions of mass–coupling parameter space have been ruled out — that appear in a large fraction of papers in the field [5, 4, 6]. The AxionLimits repository [5] tracks over 600 limits and 160 projected sensitivities, yet is maintained by a single volunteer who must identify new results, extract data from figures or tables, convert between parameterisations (e.g. kinetic mixing χ vs. coupling g [4]), write plotting code, and regenerate figures. With multiple relevant papers appearing weekly, this concentrates critical community infrastructure on a single point of failure.

*This paper was drafted with the assistance of large language models (Claude) under human supervision and editorial control.

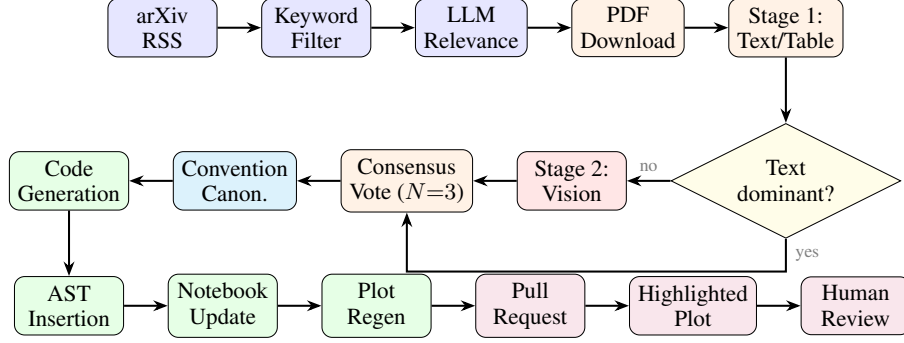


Figure 1: Pipeline architecture. Papers flow from arXiv through keyword and LLM filtering, two-stage extraction (text-first, vision fallback), multi-read consensus voting, physics convention canonicalization, code generation with AST-based insertion, plot regeneration, and finally a pull request with a highlighted constraint plot for human review.

We present a pipeline that partially automates this workflow: (1) **Discovery** via daily arXiv monitoring with keyword filtering and LLM relevance classification; (2) **Extraction** using a two-stage approach (text/tables first, vision fallback for plot-only papers) with multi-read consensus voting and a physics convention-canonicalization layer; (3) **Integration** through automated code generation with Abstract Syntax Tree (AST)-based insertion, notebook updates, and plot regeneration; and (4) **Review** via PRs with highlighted constraint plots. The system also includes a weekly preprint version checker and a historical backfill workflow using INSPIRE-HEP [7].¹

Related work. Chart understanding benchmarks (ChartQA [9], PlotQA [10]) address visual QA but not numerical extraction from dense log–log exclusion plots; document parsers like GROBID [11] handle metadata, not figure digitisation. To our knowledge, no existing system automates the full loop from discovery through extraction, code integration, and human review for a live scientific repository.

Contributions. (i) An end-to-end pipeline that turns an arXiv paper into a reviewable code change for a real constraint repository. (ii) An extraction stack that treats LLM reads as noisy samples and reconciles them by consensus with source-quality tiering, and pairs them with a physics convention-canonicalization layer whose conversions were established by a guided physics-derivation assistant rather than by prompting. (iii) A 346-paper benchmark and scoring methodology that grounds truth in the upstream repository and separates extraction error from non-comparability. (iv) The empirical finding that, with these in place, log–log plot reading is no longer the dominant error source — the frontier shifts to deterministic, convention-bound rare coupling types.

2 System Architecture

Figure 1 shows the end-to-end pipeline.

2.1 Two-Stage Extraction with Consensus Voting

The extraction pipeline prioritises text and table data (Stage 1) over vision-based figure reading (Stage 2), motivated by both cost and accuracy.

Stage 1: Text and tables. The PDF is parsed with PyMuPDF [12] and the full text sent to an LLM [8] with a structured prompt requesting JSON output: coupling type, mass–coupling data points, the model’s declared units/convention, and metadata. Prompt-injection defences strip control characters and wrap content in sentinel delimiters.

Stage 2: Vision fallback. Vision is invoked only when Stage 1 is not *clearly dominant* (a valid mass window, ≥ 5 points, sufficient confidence, non-degenerate). Selected figure pages are rendered to images; the model reads axis labels, tick marks, and curve positions from log–log exclusion plots.

¹Code, evaluation benchmark, and prompts: <https://github.com/FaroutYLq/AutoAxionLimits>

Consensus across noisy reads. LLM extraction is non-deterministic, so each paper is read several times independently ($N=3$) and reconciled. The coupling type is chosen by majority vote; the reported curve is the *medoid* — the candidate with the smallest median pairwise distance to the others in log–log space. The medoid is a real sample, not an average, so consensus rejects one-off mis-reads without ever fabricating a non-physical curve by averaging incompatible ones.

Quality-tier source selection. When candidates disagree, they are ranked by a lexicographic quality tuple rather than by raw point count: validity (values in physically allowed ranges), non-degeneracy (a traced curve must span ≥ 1 dex, not a flat artefact), recoverability, *source tier* (table > text > figure–vision), corroboration between reads, and confidence. A sparse text or table snippet (≤ 3 points, e.g. a headline bound quoted in prose) is demoted below a traceable multi-point figure curve, so a single quoted number cannot override an actual digitised constraint.

2.2 Convention Canonicalization

The same physics is reported in incompatible y -variables across papers and even across files within one coupling type, and the conversion often lives in the repository’s plotting code, not in the paper. Examples we encountered: axion–nucleon couplings stored as the derivative coupling $g_{aNN} = C_N/2f_a$ [GeV $^{-1}$] but plotted dimensionless after an in-code $\times 2m_N$ (with a per-file $\times m_N$ exception for one detector); scalar/dilaton bounds stored as a genuine Damour–Donoghue $d_e \sim 10^{-3}$ in some files but as a fifth-force strength $\alpha \sim 10^7$ in others (related by $d_e = Q^{-1}\sqrt{\alpha}$, a 10–15 dex *apparent* offset that is not an error); dark-photon kinetic mixing reported as ε^2 vs. $\chi = \varepsilon$; and large sentinel values ($10^{20}, 10^{30}, 10^{99}$) that are `fill_between` polygon-closing vertices, not data.

These are *deterministic* mismatches: no amount of prompting fixes them, because the conversion depends on per-file repository storage that the paper never states. We encode them in a per-file convention registry that the comparator (and the production pipeline) applies to canonicalize both the extracted and reference curves before any residual is computed; values it cannot map are flagged [CONVENTION REVIEW] with capped confidence rather than emitted as a silent wrong number. Crucially, the conversion factors were not guessed: they were derived with a guided physics-derivation assistant (GPD), which produced a reference that is *code-verified* against the repository’s plotting source and notebooks and *citation-audited* against the literature. Getting these right is a small but exacting physics task, and treating it as one — rather than as a prompting problem — is what made the registry trustworthy.

2.3 Code Generation, Integration, and Review

The LLM generates a `@staticmethod` for the relevant plotting class (a post-generation guard enforces the decorator); insertion uses Python’s AST module to locate the last method in the target class — never regex — preventing corruption. Notebook cells are inserted via `nbformat` and plots regenerated headlessly via `nbconvert`. Every update becomes a pull request, with low-confidence or unresolved-convention extractions flagged in the title. Crucially, we render a “highlighted” constraint plot in which all existing limits are greyed out and only the new proposed limit is shown in red (Figure 2), enabling rapid visual verification by domain experts without reading code.

2.4 Workflows and Cost

The **daily arXiv digest** queries, filters, classifies relevance, and processes matching papers. The **weekly preprint checker** scans existing data files for arXiv IDs and re-extracts on version updates to catch changed results; one PR detected that a JWST dark photon search [13] had changed a limit to a projection upon publication. The **historical backfill** searches INSPIRE-HEP [7] for older papers by citation count, splitting large jobs across daily runs.

The pipeline runs on a frontier model (Claude Opus [8]) to maximise extraction quality. With $N=3$ consensus and conditional vision, a paper triggers roughly five to nine model calls; text-only papers (the majority) avoid expensive image tokens. We do not present this as a cheap pipeline — per-paper cost is on the order of tens of cents to a few dollars, dominated by vision papers — but it remains far below the expert time it substitutes for, and text-first gating bounds the spend.

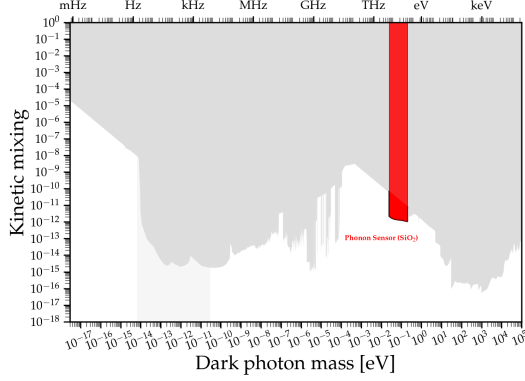


Figure 2: Highlighted dark photon constraint projection plot from a production pull request for a phonon-sensor proposal [14]. Existing limits in grey; new proposal in red. This format lets reviewers verify the proposal visually.

Table 1: Benchmark performance on 346 papers. Residual statistics are over the 271 comparable papers (243 with mass-range overlap); $0.3 \text{ dex} \approx$ a factor of two. The right block reports the median residual broken out by the winning extraction source (text vs. figure-vision) and by how per-paper residuals are pooled: the *micro*-average weights every paper equally, the *macro*-average weights every coupling type equally (two example types shown).

Aggregate metric	Value	By source / pooling	Med. resid.
Coupling-type accuracy	90.5%	Text ($N=148$)	0.326 dex
Median residual	0.331 dex	Figure-vision ($N=119$)	0.338 dex
Within factor 2	48.4%	Micro-avg (per paper)	0.331 dex
Within factor 3	61.7%	Macro-avg (per type)	1.115 dex
Mean interp. coverage	76.2%	AxionPhoton ($N=127$)	0.207 dex
Zero-overlap papers	10.3%	AxionEDM ($N=3$)	4.649 dex

3 Evaluation

Benchmark and ground truth. We evaluate on 346 papers spanning 14 coupling types, selected as all curated data files [5] with a known arXiv ID. The ground truth for each is the upstream repository’s own digitised curve — never data digitised by our own model — which avoids circularity but carries a consequence: the reference was itself digitised and rescaled from the same papers, so a *perfect* extraction still shows a nonzero residual equal to the upstream digitisation/convention gap. Our reported residuals are therefore an *upper bound* on true extraction error.

Metric. We compare in $\log_{10}$ - $\log_{10}$ space: a log-log interpolation is built from the extracted points and evaluated at ground-truth masses, giving a per-point residual $|\log_{10} g_{\text{ext}} - \log_{10} g_{\text{truth}}|$ ($0.3 \text{ dex} \approx$ factor of 2). We report median residual and *interpolation coverage* (fraction of GT masses the extracted curve spans) separately, because a curve can be accurate where it overlaps yet cover the wrong mass window. A reverse pass (GT interpolated onto the extracted masses) flags curves that run past the true mass range.

Honest disposition taxonomy. A curve is scored only against a same-coupling GT curve, so non-comparable cases (wrong coupling, incompatible convention, single-point or unusable GT, no extracted points) are tracked separately rather than polluting the residual statistics. Of 346 papers, 271 are comparable; the statistics below are over those.

Classification labels. Coupling type is scored against the repository’s directory structure. The property labels (*is new limit*, *is projection*, *data source*) are scored against an *independent* LLM labeler (a different model and prompt from the extractor it grades), making this a cross-model check rather than self-agreement; a 15-paper human audit found labeler-human agreement of 15/15, 15/15, and 14/15 on the three fields.

Results. Table 1 summarises the benchmark. Coupling-type classification is reliable (90.5%); 90% of comparable papers achieve mass-range overlap, with a median residual of 0.33 dex and 48% of curves within a factor of two, at 76% mean coverage. The reverse pass agrees (median 0.34 dex), indicating the curves cover the right mass extent, not just the right values where they overlap.

The most consequential result is the *source breakdown*: figure-vision extraction (0.338 dex, 46.8% within factor two) is now on par with text extraction (0.326 dex, 49.7%). Reading dense log–log plots — historically the hard part of this task — is no longer the limiting factor; consensus voting and quality-tier selection over multiple reads close most of the figure-vision gap.

Where the difficulty now lives. The frontier has shifted to *rare coupling types*. The per-paper micro-average (0.33 dex) is dominated by the most common type (AxionPhoton, 0.21 dex, $N=127$); the per-type macro-average is 1.1 dex. Common types are strong (DarkPhoton 0.41, AxionElectron 0.38), but small-sample, convention-heavy types lag: AxionNeutron 0.77, VectorBL 0.91, AxionMass 0.94, and AxionEDM 4.6 dex. AxionEDM is the clearest case: the operator coupling $g_{a\gamma\gamma}$ is flat in mass while the experimentally reported *response* $\propto 1/m_a$ folds in the field amplitude — a convention our registry flags but does not yet auto-convert. These types also concentrate the coupling-type misclassifications, so they compound: a wrong type is scored against the wrong reference.

4 Lessons Learned

The worst errors are deterministic physics, not perception. The largest residuals were never about pixels but unit/convention mismatches — the axion–nucleon $2m_N$ factor, the scalar d_e vs. $\sqrt{\alpha}$ 10–15 dex gap, dark-photon ε vs. ε^2 , sentinel values mistaken for data — that are fixed, not stochastic, and depend on per-file repository conventions absent from the paper. No prompt fixes them; the high-leverage fix was a separate, auditable physics layer. We used a guided physics-derivation assistant (GPD) to produce a *code-verified, citation-audited* per-file convention registry, with a [CONVENTION REVIEW] flag for anything outside the verified set. The general lesson: automated scientific extraction couples a *stochastic* perception problem (reading plots, best handled by sampling and voting) with a *deterministic* domain-knowledge problem (conventions, best handled by verified derivation), and the two demand different tools.

Score honestly. Grounding truth in the upstream repository (never in our own model’s digitisation) avoids circularity and makes the residual a defensible upper bound. Separating non-comparability (wrong-coupling, point-reference, convention-mismatch) from genuine error keeps the headline numbers meaningful, and scoring property labels against an independent LLM labeler rather than placeholders restores those metrics as a fair cross-model test.

Engineering that paid off. Text-first extraction with conditional vision keeps the majority of papers off the expensive vision path; AST-based insertion produces syntactically valid methods; and highlighted plots make review accessible to non-coders.

What is still hard. Rare coupling types remain the open problem: small samples, idiosyncratic conventions (the AxionEDM operator-vs-response case, scalar fifth-force files), and the misclassifications that send them to the wrong reference. At 90% coupling accuracy roughly one proposal in ten still targets the wrong plot, so human-in-the-loop review is essential, not optional. A residual $\sim 10\%$ of papers also miss the mass window entirely (too few points or the wrong panel).

Governance. The pipeline is deployed and has generated limit proposals; none have merged, because no review process for AI-generated scientific data exists. We recommend a community editorial board analogous to PDG reviewers [15], where collaborations validate their own limits. More fundamentally, publishing machine-readable limit data alongside papers would bypass the extraction bottleneck altogether — the highest-leverage change the field could make.

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